



Cambridge International AS & A Level

CANDIDATE
NAME

Downloaded from <https://www.cambridge.org/core>. University of Cambridge, on 01 Jun 2018 at 12:00:00, subject to the Cambridge Core terms of use, available at <https://www.cambridge.org/core/terms>. <https://doi.org/10.1017/9781315326478.008>

CENTRE
NUMBER

--	--	--	--	--

CANDIDATE
NUMBER

--	--	--	--

FURTHER MATHEMATICS

9231/23

Paper 2 Further Pure Mathematics 2

May/June 2026

2 hours

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].

This document has **20** pages. Any blank pages are indicated.



DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN



[5]

[illegible]



- 3** Let $I_n = \int_0^a (\ln(x+1))^n dx$, where n is a non-negative integer and $a > 0$.

- (a) By considering the derivative of $(x+1)(\ln(x+1))^n$, or otherwise, show that for $n \geq 1$

$$I_n = (a+1)(\ln(a+1))^n - nI_{n-1}.$$

[3]





[3]

[Turn over





4 Find the particular solution of the differential equation

$$49 \frac{d^2x}{dt^2} + 14 \frac{dx}{dt} + x = 196e^{\frac{1}{7}t},$$

given that, when $t = 0$, $x = 7$ and $\frac{dx}{dt} = 0$.

[10]





DO NOT WRITE IN THIS MARGIN

- 5 (a) Starting from the definitions of sech and $\tanh$ in terms of exponentials, prove that

$$1 - \operatorname{sech}^2 t = \tanh^2 t.$$

[3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

It is given that

$$x = t \sinh t \quad \text{and} \quad y = t \cosh t.$$

- (b) Show that $\frac{dy}{dx} = \frac{t \tanh t + 1}{t + \tanh t}$.

[3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....





(c) Show that $\frac{d^2y}{dx^2} = \frac{(t^2 - 2)\operatorname{sech}^n t}{(t + \tanh t)^n}$, where n is an integer to be determined.

[6]





6 (a) Show that an appropriate integrating factor for

$$(1-x^2)\frac{dy}{dx} + y = (1-x^2)\sqrt{1-x}$$

is $\sqrt{\frac{1+x}{1-x}}$.

[4]

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN





(b) Hence find the solution of the differential equation

$$(1-x^2)\frac{dy}{dx} + y = (1-x^2)\sqrt{1-x}$$

for which $y = 1$ when $x = 0$.

[6]



- 7 (a) Use the substitution $u = 1 - x^2$ to find $\int \frac{x}{\sqrt{1-x^2}} dx$. [3]

.....

.....

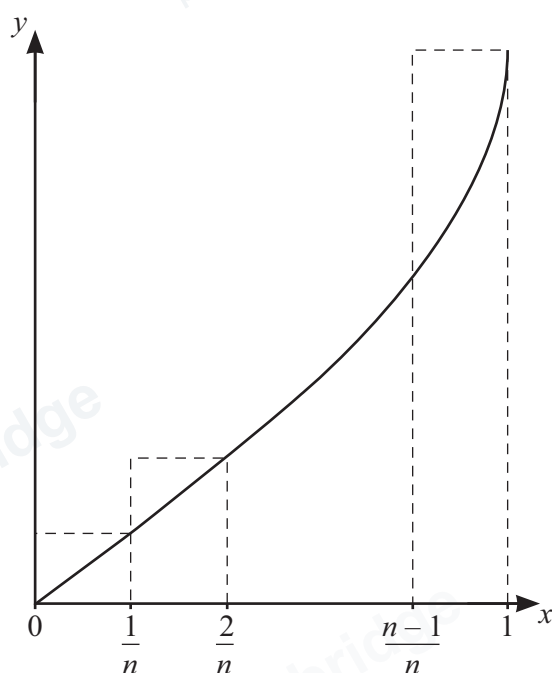
.....

.....

.....

.....

.....



The diagram shows the curve with equation for $y = \sin^{-1}x$ for $0 \leq x \leq 1$, together with a set of n rectangles of width $\frac{1}{n}$.

- (b) By considering the sum of the areas of these rectangles, show that

$$\sum_{r=1}^n \frac{1}{n} \sin^{-1}\left(\frac{r}{n}\right) \geq \frac{1}{2}\pi - 1. \quad [5]$$

.....

.....

.....

.....

.....





PapaCambridge
papacambridge.com

PapaCambridge
papacambridge.com

- (c) Use a similar method to find, in terms of n , an upper bound for $\sum_{r=1}^n \frac{1}{n} \sin^{-1}\left(\frac{r}{n}\right)$. [3]

- (d) Deduce the exact value of $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n} \sin^{-1}\left(\frac{r}{n}\right)$. [1]





- 8 (a) The constants a and b are such that the system of equations

$$6x - 12y - 12z = 1,$$

$$ax + 4y + z = 2,$$

$$2x + by - 8z = 3,$$

does not have a unique solution. Find a in terms of b .

[3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

- (b) Given that $a = -\frac{1}{4}$ and $b = -32$, show that the system of equations in part (a) is inconsistent. Interpret this situation geometrically.

[3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....





The matrix $\mathbf{A}$ is given by

$$\mathbf{A} = \begin{pmatrix} 6 & -12 & -12 \\ -1 & 4 & 1 \\ 2 & -8 & -8 \end{pmatrix},$$

- (c) Show that $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ is an eigenvector of $\mathbf{A}$ and state the corresponding eigenvalue. [2]

.....

.....

.....

.....

.....

.....

- (d) Find a matrix $\mathbf{P}$ and a diagonal matrix $\mathbf{D}$ such that $\mathbf{A} = \mathbf{PDP}^{-1}$. [7]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....





PapaCambridge
papacambridge.com

PapaCambridge
papacambridge.com

.....

.....

PapaCambridge
papacambridge.com

PapaCambridge
papacambridge.com

DO NOT WRITE IN THIS MARGIN



If you use the following page to complete the answer to any question, the question number must be clearly shown.

[illegible]





DO NOT WRITE IN THIS MARGIN





Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (Cambridge University Press & Assessment) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in our Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge International Education is the name of our awarding body and a part of Cambridge University Press & Assessment, which is a department of the University of Cambridge.

